

Internship in Physical Organic Chemistry

Week 1 — Part 1: Setting up your working environment

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Goals:

- Install and configure Zotero as your reference manager
- Set up your Overleaf project and learn basic LaTeX
- Share your Overleaf project with your supervisor

This week you will begin working as a researcher. You will learn how to find relevant scientific papers, how to read and extract information from them, and how to build a scientific argument.

At the end of this week, you will submit a complete draft of your Introduction in Overleaf.

Before starting the literature work, you must set up a working environment that will be used throughout the entire internship. This environment has two components: a system to manage scientific papers and a system to write your report. If this is not done correctly at the beginning, the rest of the work will become disorganized and difficult to maintain.

1 Setting up your reference manager

A reference manager is a software that allows you to collect, store, and organize scientific papers. In research, you will typically work with many articles at the same time, and it is essential to keep track of where information comes from. Your reference manager acts as a central database: every paper you read, use, or cite will be stored there, together with its bibliographic information. Later, when writing your report, the reference manager allows you to retrieve and cite these papers correctly

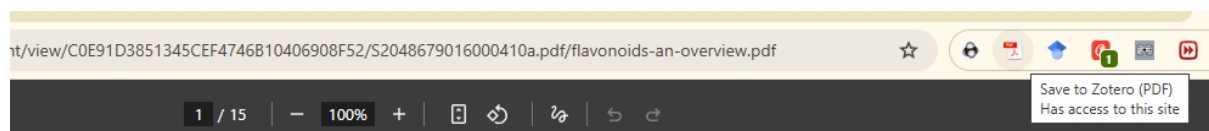
1.1 Installing Zotero

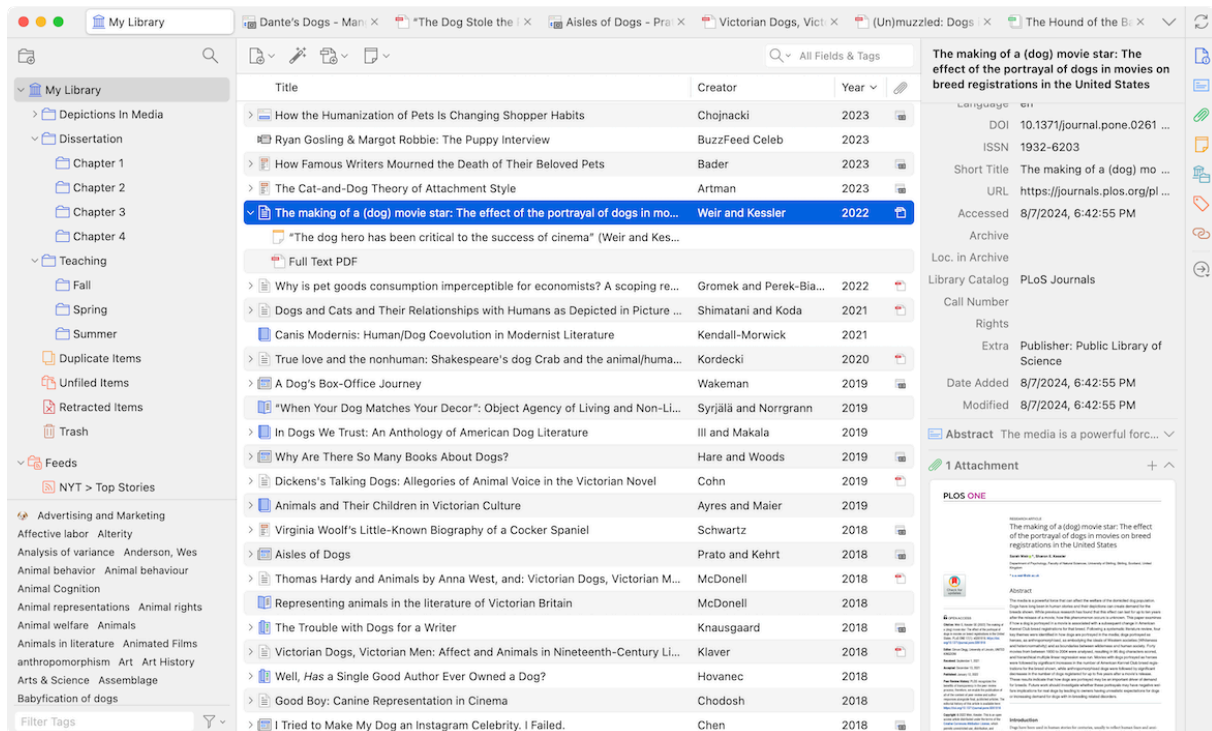
You will use Zotero as your reference manager. Go to <https://www.zotero.org>, download the software, and install it. Create an account and log in inside the application.

You must also install the Zotero browser connector from the same website:

<https://www.zotero.org/download/> This connector allows you to save papers directly from your browser without manually downloading and renaming files.

Once installed, open Zotero and your web browser. You should now see the Zotero application window and, in your browser toolbar, a small Zotero icon. This icon changes depending on the type of content you are viewing (article, PDF, webpage):





1.2 Importing a paper

To test that everything is working, go to Google Scholar <https://scholar.google.com/> and search for “flavonoid MS/MS fragmentation”. Open one of the results that leads to a scientific article. Click on the Zotero icon in your browser. The paper should immediately appear in your Zotero library.

At this point, you must verify that the entry is correct. Select the paper in Zotero and look at the information displayed: the title, authors, journal, and year must be present. In addition, a PDF file should be attached. If any information is missing or incorrect, you must edit it manually. Zotero does not always import data perfectly, and incorrect metadata will lead to incorrect citations later.

Create a new collection in Zotero called “week1_literature” and move your imported papers into this collection. This ensures that all papers used for this project are grouped together and can be accessed easily.

1.3 Storage

Your Zotero library is part of your research data and must not be lost. When working with Zotero, it is important to understand that two different types of information are stored: the bibliographic data (titles, authors, journal, year) and the PDF files attached to each reference.

Zotero provides a built-in synchronization system that allows you to access your library from different computers. By default, all bibliographic data is synchronized online without limitation. This means that your references will always be available once you log in to your account. However, storage for PDF files is limited to 300 MB in the free version. This is sufficient for a small number of papers, but you should be aware of this limit.

To activate synchronization, open Zotero and go to Edit → Preferences → Sync. Enter your Zotero account credentials and enable synchronization. Once this is done, your library will automatically update whenever changes are made.

For this internship, you Zotero’s built-in synchronization will be sufficient for your needs. However, as your scientific library will likely grow over time, you will quickly exceed the free

quota, for which you may consider configuring external storage systems such as [Dropbox for Zotero](#).

Regardless of how you choose to store your pdfs inside Zotero, it is important that all of them are managed in one place. You should always open PDFs through Zotero and not from your file system. Do not move, rename, or modify files manually outside the application, as this can break the link between the reference and the PDF.

After enabling synchronization, restart Zotero and verify that your papers are still present and that PDFs open correctly. If anything is missing or does not work, the setup must be corrected before continuing.

2 Setting up your writing environment

Overleaf is an online editor used to write documents in LaTeX. During your internship, you will use it to write your report from the beginning. Unlike a word processor, LaTeX separates content from formatting. You write structured text, and the final document is generated automatically.

LaTeX is widely used in scientific environments because it produces consistent, high-quality documents and handles complex structures such as equations, references, and figures reliably. Unlike word processors such as Microsoft Word, LaTeX does not rely on manual formatting. This reduces formatting errors and ensures that documents remain consistent even as they grow.

In addition, LaTeX integrates naturally with reference managers and allows precise control over citations and bibliographies. This is essential for scientific writing, where every statement must be properly referenced.

For these reasons, LaTeX is the standard in many areas of chemistry, physics, and mathematics.

2.1 Account and access

Go to <https://www.overleaf.com> and create an account. The free version is sufficient for this internship and allows you to create and compile LaTeX documents directly in your browser without installing any software.

Once logged in, you can create new projects and access them from any computer. This ensures that your work is always available and does not depend on a local installation.

2.2 Creating a project

Create a new project and name it:

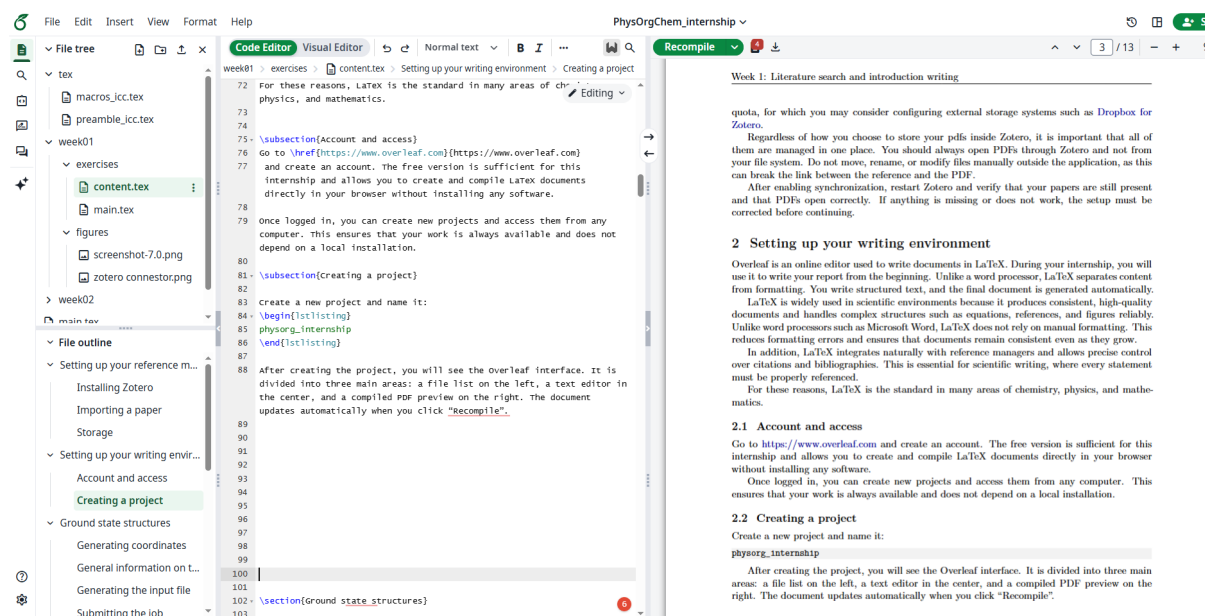
```
physorg_internship
```

After creating the project, you will see the Overleaf interface. It is divided into three main areas: a file list on the left, a text editor in the center, and a compiled PDF preview on the right. The document updates automatically when you click “Recompile”.

Open the main file (usually `main.tex`) and add the following sections:

```
\section{Introduction}
\section{Background}
\section{Experimental}
\section{Results and Discussion}
\section{Conclusion}
```

Click “Recompile”. The PDF on the right should display these section titles. If it does not, correct the errors before continuing.



2.3 Templates

Overleaf provides templates for scientific documents. You can access them directly from the Overleaf interface. Templates are useful because they already contain the structure and formatting expected in scientific writing.

For this internship, you may use one of the report templates available here:
<https://www.overleaf.com/latex/examples/tagged/report/>.

2.4 LaTeX primer

You do not need prior knowledge of LaTeX to start using Overleaf. However, you must understand how a LaTeX document is structured and how to perform basic operations while writing.

A LaTeX document is written as plain text using commands. These commands define the structure of the document. The document is then compiled to produce a PDF.

In Overleaf, you write in the central editor and press Recompile to update the PDF.

A minimal LaTeX document has the following structure:

```
\documentclass{article}
\begin{document}
Your text goes here.
\end{document}
```

All content must be written between `\begin{document}` and `\end{document}`.

In practice, you will use a small number of commands repeatedly. The most important ones are shown below.

Command	Description
<code>\section{Title}</code>	Creates a section
<code>\subsection{Title}</code>	Creates a subsection
<code>\textbf{text}</code>	Bold text
<code>\emph{text}</code>	Emphasized text
<code>\\</code>	Line break
<code>\cite{key}</code>	Insert citation
<code>\ref{label}</code>	Reference figure or section
<code>%</code>	Comment (ignored during compilation)

Text is written normally between commands. For example:

```
\section{Introduction}
```

This is your introduction. You can write full sentences here.

Figures can be added using the following structure:

```
\begin{figure}  
\centering  
\includegraphics[width=0.5\textwidth]{filename}  
\caption{Caption text}  
\end{figure}
```

The image file must be uploaded to your Overleaf project before it can be used.

You are not expected to learn LaTeX in full. In practice, you will reuse existing code, modify it, and compile again. This is the standard workflow when writing scientific documents in LaTeX.

After you have finished setting up your LaTeX environment, share your project with me and provide me with editing rights.